P-Nav

Technical Manual

Android GNSS Positioning and Observation Quality Analysis Platform

Version 1.0

Item	Details
Software Version	v1.0.0
Manual Version	v1.0
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Technical Support	lglu@ecut.edu.cn
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1.Overview

P-Nav is an integrated software platform that turns an Android smartphone into a GNSS measurement and positioning tool. It acquires raw GNSS measurements—including pseudorange, carrier phase, Doppler shift, and carrier-to-noise-density ratio (C/N_0)—through the Android GNSS APIs. The application supports three positioning modes, performs epoch-by-epoch GNSS observation quality analysis, and exports data in standard formats for subsequent post-processing.

Capability	Description
Positioning Modes	Single Point Positioning (SPP), Differential GNSS (DGNSS), and Real-Time Kinematic (RTK) positioning
Supported Constellations	GPS, GLONASS, BeiDou (BDS), Galileo, and QZSS
Supported Signals and Frequencies	GPS L1/L2/L5; BDS B1I/B1C/B2b/B2a; Galileo E1/E5a/E5b; GLONASS G1/G2
Observation Quality Analysis	Multipath-related indicators (MP1/MP2/MP5), cycle-slip detection, carrier-phase noise, C/N_0, and ionospheric delay rate
Data Export	RINEX 3.04 observation files and GnssLogger-compatible CSV files

2.Installation and First Launch

System Requirement: Android 9 (API 28) or later.

1. Transfer P-Nav-v1.0.0-debug.apk to your Android smartphone.
2. Enable permission to install apps from unknown sources for the app used to open the APK file, then install P-Nav.
3. On the first launch, a language selection dialog appears. Select Chinese or English. The interface switches to the selected language immediately, and the selection dialog will not appear again on subsequent launches.
4. Grant the requested location permission (required) and storage-related permissions, if requested, for data export.

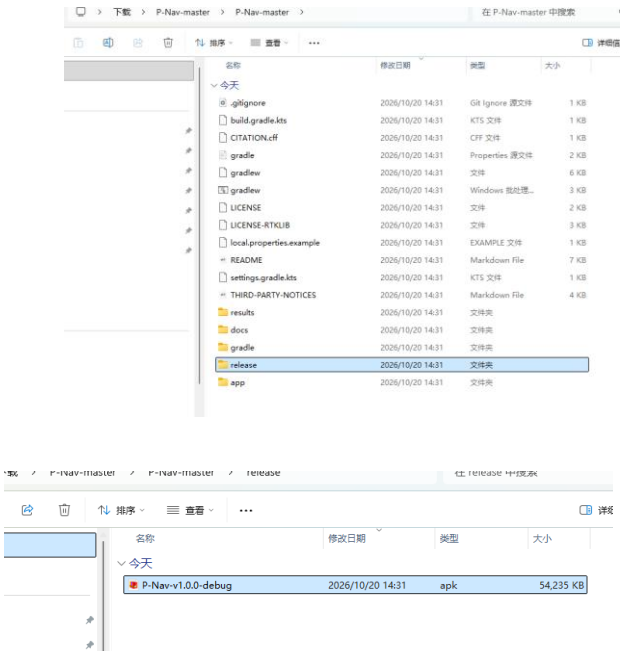


Figure 2-1 APK files

Note: If an older version signed with a different certificate is already installed, uninstall it before installing this version.

3. Interface and Workflow

P-Nav provides five pages accessible through the bottom navigation bar: Data Collection, Satellite Status, Quality Analysis, Positioning, and Settings.

3.1 Typical Workflow

A typical workflow for GNSS data collection, observation assessment, and positioning is as follows:

1. Data Collection: Start recording GNSS observations and check the measurement rate and signal strength.
2. Satellite Status: Inspect satellite visibility and geometry to assess the current observation conditions.
3. Quality Analysis: Evaluate GNSS observation quality to determine whether the measurements are suitable for high-precision positioning.
4. Positioning: Configure the correction source and positioning mode, then monitor the positioning results.
5. Data Export: Stop data collection and export the recorded observations in RINEX or CSV format.

All these operations are integrated into a single application, allowing users to collect data, assess observation quality, perform positioning, and export data without switching to external software.

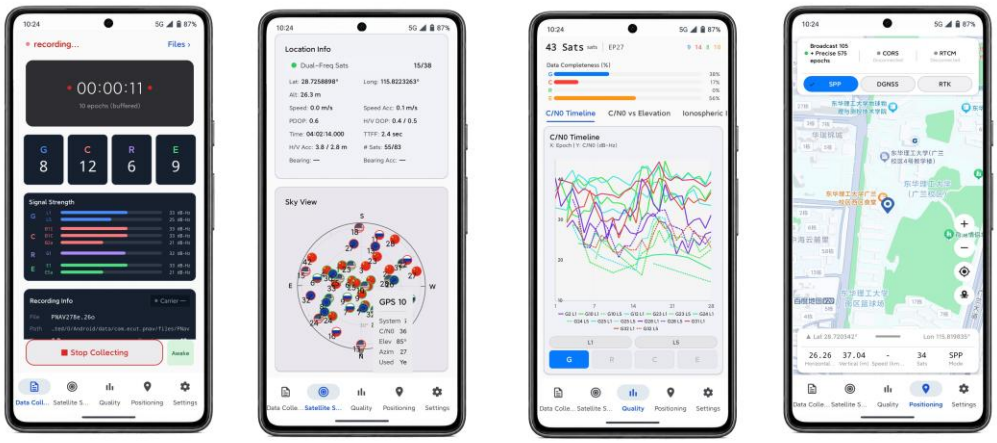


Figure 3-1 Main Interface

4. Data Collection Page

The page is divided into five sections arranged from top to bottom.

4.1 Status Bar and File Access

The top of the page displays the current recording status. recording... indicates that data collection is in progress, while Ready indicates that the application is idle. Tap **Files ›** on the right to open the list of recorded files, where previous sessions can be viewed, deleted, or used for subsequent operations.

4.2 Timer and Epoch Count

The large timer displays the elapsed recording time for the current session. The smaller text below it shows the number of epochs currently buffered in memory.

At a sampling rate of 1 Hz, the epoch count should be approximately equal to the elapsed time in seconds (e.g., 11 seconds corresponds to approximately 10–11 epochs). A significantly lower count may indicate that GNSS measurement callbacks are not being received as expected. Common causes include incomplete location permissions or power-saving restrictions on background processing.

After data collection stops, the buffered epoch count remains displayed in the timer area, allowing users to verify the amount of data collected during the session.

4.3 Satellite Counts by Constellation

Four tiles show the satellite counts for GPS (G), BeiDou (C), GLONASS (R), and Galileo (E), respectively. The displayed count represents the number of satellites from each constellation currently used in the positioning solution.

- In open-sky environments, each constellation typically provides 6–12 satellites, although the actual number depends on satellite visibility, device capabilities, and the observation conditions.
- If a constellation remains at zero for an extended period, the device may not support the corresponding signals, or the signals may be obstructed.
- Frequent fluctuations in the counts may indicate intermittent satellite tracking or loss of lock.

4.4 Signal Strength

The signal-strength chart displays the carrier-to-noise-density ratio (C/N_0) for individual satellites and frequencies. Bars are color-coded by constellation. Each row identifies the signal frequency on the left (e.g., L1, L5, B1I, B1C, B2a, G1, E1, or E5a), with the corresponding C/N_0 value in dB-Hz on the right.

The chart helps users identify potential signal-quality issues **during data collection**.

Observed Condition	Possible Interpretation
An entire frequency row is missing	The device may not track that frequency.
One satellite has a substantially lower C/N_0 than other satellites in the same constellation	The signal may be obstructed in that satellite's direction.
C/N_0 values decrease across many satellites	The device may have entered an obstructed environment, or the antenna orientation may have changed.
C/N_0 falls below 20 dB-Hz	The signal is weak and may be unsuitable for reliable high-precision positioning.

4.5 Collection Information

This section displays the current session's file name, storage path, file size, recording duration, and carrier-phase availability.

Field	Description
File	Name of the RINEX observation file generated for the current session, following the format PNAV<DOY><session-letter>.<two-digit-year>o.
Path	Directory where the file is stored, located in the application's private storage area.
Size	Current file size. The file is written after data collection stops, so the displayed size may remain zero or very small during recording.
Duration	Elapsed recording time for the current session.
Carrier	Indicates carrier-phase availability. Carrier ✓ means that valid carrier-phase observations have been reported by the device. Carrier — means that carrier-phase observations are unavailable, in which case RTK positioning cannot operate.

4.6 Control Buttons

- **Stop Collecting:** Stops data collection and generates the RINEX observation file.
- **Awake:** Keeps the screen on, which is useful during long static observation sessions.

Important: During data collection, GNSS observations are buffered in memory and written to the RINEX file only when collection stops. Longer sessions therefore require more time to finalize and write the file. Wait for processing to finish, and do not force-close the application. Force-closing it may result in the loss of all observations collected during the current session.

5. Satellite Status Page

5.1 Positioning Information Card

This card provides a summary of the current positioning solution and its status.

Field	Description
Dual-Freq Sats	Number of satellites observed on two frequencies / total number of visible satellites.
Lat / Long	Latitude and longitude of the current positioning solution.
Alt	Ellipsoidal height (WGS 84), not elevation above mean sea level.
Speed / Speed Acc	Doppler-derived speed and its estimated accuracy.
PDOP	Position Dilution of Precision, a measure of satellite geometry.
H/V DOP	Horizontal / vertical Dilution of Precision.
Time	Timestamp of the current positioning solution.
TTFF	Time to First Fix.
H/V Acc	Estimated horizontal / vertical positioning accuracy.
# Sats	Number of satellites used in the positioning solution / total number of tracked satellites.
Bearing / Bearing Acc	Heading and its estimated accuracy.

Interpretation Guidelines

- **PDOP below 2** generally indicates favorable satellite geometry; **2–5** indicates moderate geometry; and **above 5** indicates poor geometry. Even when many satellites are visible, poor geometry can significantly degrade positioning accuracy.
- A large difference between the numerator and denominator of **# Sats** indicates that many tracked satellites may have been excluded from the positioning solution by quality-control filters, for example, because of low elevation angles, incomplete signal synchronization, or insufficient C/N₀. Check the Quality Analysis page to investigate potential causes.

5.2 Skyplot

The skyplot displays satellite positions in the sky using polar coordinates.

- **Angular position:** Represents azimuth, with north (N), east (E), south (S), and west (W) indicating the cardinal directions.
- **Radial distance from the center:** Represents elevation angle. The center corresponds to the zenith (90°), while the outer circle corresponds to the horizon (0°).
- **Colors and icons:** Identify satellite constellations. Flag icons facilitate quick constellation identification.

Tap any satellite to view its details, including constellation, C/N_0, elevation angle, azimuth, and whether it is used in the positioning solution.

Interpretation Guidelines

Ideally, satellites should be distributed relatively evenly across the visible sky. If they are concentrated on one side, buildings or other obstacles may be blocking signals from the opposite direction. This can degrade satellite geometry, increase PDOP, and reduce positioning accuracy. Satellites at low elevation angles, near the outer edge of the skyplot, are generally more susceptible to multipath and atmospheric effects.

6. Quality Analysis Page

The Quality Analysis page is one of P-Nav's core features. All metrics are computed epoch by epoch and can be monitored in real time during data collection.

6.0 Page Overview

The page is organized into three sections from top to bottom:

- **Top Statistics Bar:** Displays the total number of satellites, the current epoch index, and satellite counts by constellation, with each constellation indicated by a distinct color.
- **Data Completeness:** Displays the proportion of expected observation epochs for which measurements were actually received, broken down by constellation. A low percentage may indicate poor satellite visibility or frequent loss of tracking.
- **Chart Area:** Provides three selectable tabs: C/N_0 Time Series, C/N_0 versus Elevation Angle, and Ionospheric Indicators. Frequency filters (L1 / L5) and constellation filters (G / R / C / E) are available below the charts.

6.1 Data Completeness

Data completeness is defined as the proportion of expected observation epochs for which measurements were actually received, calculated separately for each constellation. A low value may indicate poor satellite visibility or frequent loss of tracking.

6.2 Carrier-to-Noise-Density Ratio (C/N_0)

The application displays C/N_0 distributions by frequency and plots their variations over time. C/N_0 is a direct indicator of received signal strength.

C/N_0	Interpretation
> 40 dB-Hz	Good
30–40 dB-Hz	Usable
20–30 dB-Hz	Weak; more susceptible to multipath and cycle slips
< 20 dB-Hz	Very weak; generally unsuitable for reliable high-precision positioning

P-Nav also displays the relationship between **C/N_0 and satellite elevation angle**. Lower C/N_0 values at low elevation angles are generally expected. A satellite with substantially lower C/N_0 than other satellites at a similar elevation angle may indicate signal obstruction or multipath effects in that direction.

6.3 Multipath-Related Indicators: MP1 / MP5

Multipath occurs when satellite signals reflected by buildings, the ground, water, or other surfaces combine with the direct signal, introducing ranging errors. A conventional approach uses code-minus-carrier (CMC) combinations. However, smartphone carrier-phase measurements are often relative to an arbitrary initial tracking value, making their absolute values unsuitable for direct use in conventional CMC processing.

P-Nav therefore employs a **time-differenced accumulation method**. For the same satellite, consecutive epochs are differenced to obtain the following combinations:

$$\Delta MP_1 = \Delta P_1 - a_1 \Delta L_1 + a_2 \Delta L_2 \tag{1.1}$$

$$\Delta MP_2 = \Delta P_1 - c_1 \Delta L_1 + a_1 \Delta L_2 \tag{1.2}$$

where $\alpha = (f_1 / f_2)^2$, and the coefficients are defined as

$$a_1 = \frac{\alpha + 1}{\alpha - 1}, \quad a_2 = \frac{2}{\alpha - 1}, \quad c_1 = \frac{2\alpha}{\alpha - 1} \tag{1.3}$$

Differencing removes the unknown initial carrier-phase offset. Accumulating the resulting differences over a continuous tracking arc produces an MP time series. Because the accumulated values may still contain a constant offset associated with the carrier-phase ambiguity, the running mean of the current arc is subsequently removed, leaving the residuals:

$$r_i = MP_i - \overline{MP}_{\text{arc}} \tag{1.4}$$

Finally, the root-mean-square (RMS) of the residuals is calculated for each constellation:

$$MP_{\text{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} r_i^2}\,] \tag{1.5}$$

Interpretation Guidelines

MP_RMS	Interpretation
< 0.3 m	Good; generally consistent with an open-sky observation environment
0.3–0.6 m	Moderate; possible mild multipath effects
> 0.6 m	Poor; potentially significant multipath effects

Regarding MP2: Most smartphone GNSS chipsets primarily track L1+L5 or L1+L2C signals, and L2 observations may be unavailable. Consequently, the interface primarily displays **MP1 and MP5**. MP2 is also calculated internally, but its result is not meaningful when the required L2 observations are unavailable.

Implementation detail for MP5: Although L1 and L5 code measurements correspond to the same geometric range, Android may report different pseudoranges for independently tracked signal components. In the dual-frequency combination, P-Nav explicitly sets $(P_\{L5\} = P_\{L1\}$ to avoid introducing an artificial multipath term.

6.4 Cycle-Slip Detection Using Three Criteria

A cycle slip is an abrupt change in the integer cycle count of a carrier-phase measurement. It can invalidate the continuity of carrier-phase ambiguity estimates and degrade positioning performance. Because individual detection criteria can produce excessive false alarms with smartphone observations, P-Nav combines three criteria.

1. Geometry-Free (GF) Combination Difference

The geometry-free combination and its epoch-to-epoch difference are defined as

$$GF = L_1 - L_2, \qquad d_{GF} = GF_k - GF_{k-1} \tag{1.6}$$

If $|\,d_{GF}\,| > 0.30\ \text{m}$, a cycle slip is declared.

2. GF Polynomial Residual

A second-order polynomial is fitted to the most recent 10 epochs of the GF time series. The residual at the latest epoch is calculated as

$$[r_k = GF_k - GF_k] \tag{1.7}$$

If $(|r_k| > 0.05\ \mathrm{m})$, a cycle slip is declared. This criterion is intended to improve sensitivity to cycle-slip events that are less readily detected by the first criterion.

3. Recursive Melbourne–Wübbena (MW) Test

The Melbourne–Wübbena combination uses both carrier-phase and pseudorange measurements to reduce the effects of geometric range, receiver and satellite clock errors, and ionospheric delay. The wide-lane phase and narrow-lane code combinations are defined as

$$\Phi_{WL} = \frac{L_1}{\lambda_1} - \frac{L_2}{\lambda_2}, \qquad P_{NL} = \frac{f_1 P_1 + f_2 P_2}{f_1 + f_2} \tag{1.8}$$

$$MW = \Phi_{WL} - \frac{P_{NL}}{\lambda_{WL}}, \qquad \lambda_{WL} = \frac{c}{f_1 - f_2} \tag{1.9}$$

In the absence of cycle slips, the MW combination should remain approximately constant over a continuous tracking arc. P-Nav recursively estimates its mean and variance. A cycle slip is declared when the new observation deviates from the estimated mean by more than four standard deviations:

$$[|MW_k - \overline{MW}| > 4\sigma_{MW}] \tag{1.10}$$

If any of the three criteria is triggered, the ambiguity arc for that satellite is reset, including the CMC arc and the reference state used for triple-difference calculations. **The multipath arc is not reset**, for the reason explained in Section 6.6.

6.5 Measurement Noise Estimation Using Triple Differences

Following the methodology described in Chinese national standard BD 420022-2019 (Sections 6.5 and 6.6), P-Nav estimates measurement noise using triple-differenced observations. Three successive differencing operations are applied across epochs to suppress common clock and geometric-range contributions, leaving a quantity dominated by measurement noise.

The noise estimate is calculated as

$$[\sigma = \sqrt{\frac{\sum TD^2}{8N}}] \tag{1.11}$$

Pseudorange noise (TD-PR) and carrier-phase noise (TD-CP) are evaluated separately to characterize the noise levels of the respective GNSS measurements for each constellation.

6.6 CMC Detrending

The code-minus-carrier combination is defined as

$$[CMC = P - L] \tag{1.12}$$

CMC contains slowly varying contributions associated with twice the ionospheric delay, as well as a constant component related to the carrier-phase ambiguity. To separate short-term variations from the slowly varying trend, P-Nav uses a 120-epoch circular buffer to fit a **fourth-order polynomial** and calculates the residuals:

$$[CMC_{\text{resid},k} = CMC_k - CMC_k^{(4)}] \tag{1.13}$$

These residuals can be used to assess multipath-related variations and measurement noise after detrending.

6.7 Ionospheric Delay Rate (IOD)

The ionospheric delay rate is derived from the temporal change in the GF combination:

$$[IOD = \frac{d_{GF} \cdot f_2^2}{f_1^2 - f_2^2} \cdot \frac{1}{\Delta t}] \tag{1.14}$$

This indicator characterizes the rate of change of the ionospheric delay and can help assess ionospheric activity. It can also be used as an auxiliary indicator for identifying anomalous observations.

6.8 Carrier-Phase Arc Quality (CSR)

The carrier-phase arc quality indicator is defined as

$$[CSR = \frac{\text{Number of epochs}}{\text{Number of detected cycle slips}}] \tag{1.15}$$

A larger CSR generally indicates better observation continuity. For single-receiver observations without an external reference, CSR provides a convenient indicator of data usability.

6.9 Three Adaptations for Smartphone GNSS Observations

The following adaptations distinguish P-Nav's quality-analysis engine from conventional GNSS analysis software. Each addresses a characteristic of Android raw GNSS measurements.

1. The GF Threshold Test Is Not Applied to L5

Inter-frequency bias (IFB) variations between smartphone tracking channels can cause the first-order GF difference for L1/L5 observations to exceed reasonable thresholds. Applying a fixed threshold of 0.30 m can therefore produce numerous false alarms. For L5, P-Nav instead relies exclusively on the ADR_STATE status flags (RESET / CYCLE_SLIP) for cycle-slip detection.

2. Newly Acquired L2 Signals Require a Warm-Up Period

The first five epochs after a signal is reacquired may exhibit transient inconsistencies. Applying GF-based detection immediately during this period can lead to false alarms. P-Nav therefore skips GF-based cycle-slip detection during these initial five epochs.

3. Cycle Slips Do Not Reset the Multipath Arc

When a cycle slip is detected, P-Nav resets the CMC arc and the reference state for triple-difference calculations, but **deliberately preserves the multipath arc**.

The rationale is that the carrier-phase ambiguity contributes a constant offset to the accumulated MP series. The arc-mean removal described in Section 6.3 can absorb this offset. Resetting the multipath arc at a cycle slip would instead introduce an artificial step into the MP time series.

7. Positioning Page

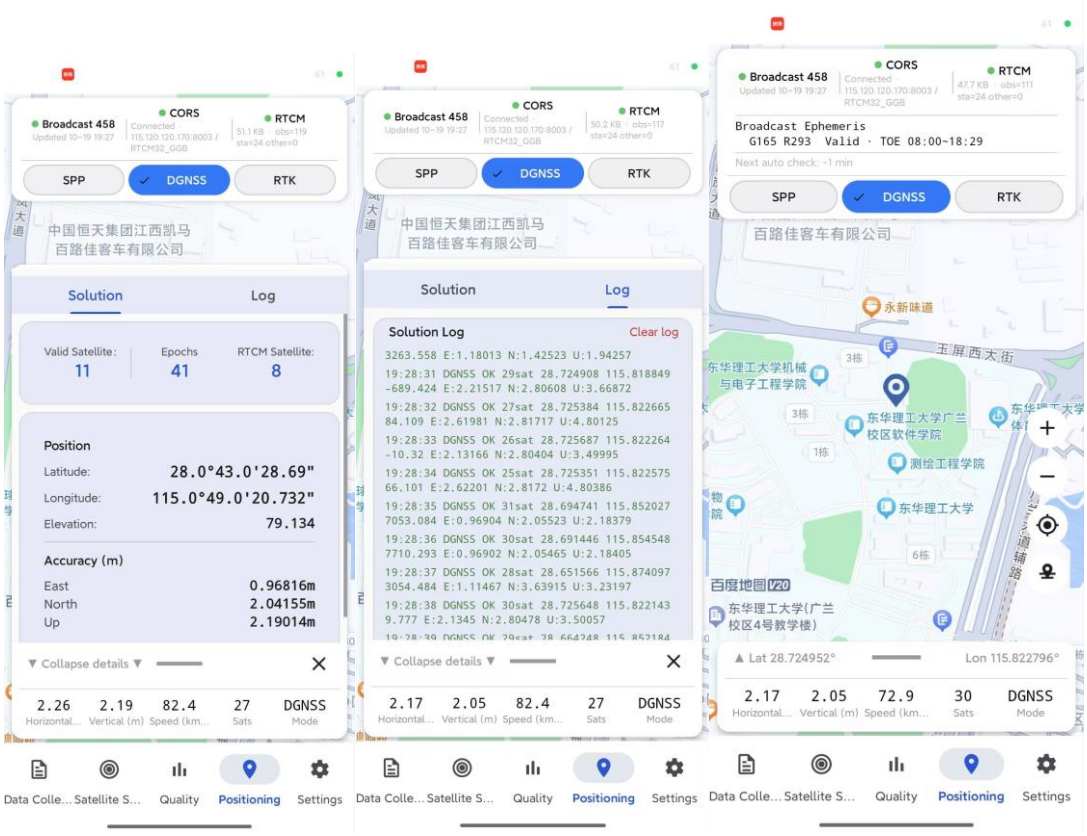


Figure 6-1 Main Interface

The page consists of three main sections: a status and mode control card at the top, an online map in the middle, and a positioning results summary at the bottom.

7.1 Status Card: Ephemerides and Correction Sources

The upper part of the card displays the status of three types of data and services.

Indicator	Description
Broadcast N	Number of broadcast ephemeris records loaded.
+ Precise N epochs	Number of precise ephemeris epochs loaded and the number of applicable observations.
CORS	A green indicator means that the CORS service is connected. A gray Disconnected indicator means that no connection is established.
NTRIP / RTCM	Controls for the two real-time correction data sources.

CORS and **RTCM** are independent controls:

- **CORS:** Connects to a network-based Continuously Operating Reference Station (CORS) service through the NTRIP protocol. Account information must first be configured on the Settings page; see Section 7.5.

- **RTCM:** Receives an RTCM data stream directly from a local base station without using an NTRIP account.

When either source provides valid correction data, RTK and DGNSS can use those corrections, provided that the incoming data are compatible with the selected positioning mode.

7.2 Positioning Mode Selection

The positioning mode can be selected using the segmented control in the lower part of the card.

Mode	Description	Requirements
SPP	Single Point Positioning using iterative weighted least squares.	Broadcast ephemerides.
DGNSS	Differential GNSS positioning using base-station pseudorange corrections.	Real-time differential correction data, typically delivered through RTCM.
RTK	Double-difference Kalman filtering with integer ambiguity resolution using the LAMBDA method and ratio testing.	Real-time differential correction data, typically delivered through RTCM.

The mode can be switched during operation, and the resulting positioning changes can be observed in real time. Switching modes reinitializes the positioning filter, so the solution may require a period of convergence before stabilizing.

7.3 Online Map

The middle section contains a map view that displays the current position in real time. Map controls include zoom in, zoom out, return to the current location, and layer selection.

The map is used **only for position visualization and does not participate in positioning calculations**. P-Nav computes positions from GNSS observations and does not use the map as a positional constraint.

If the map does not appear, a common cause is a mismatch between the Baidu Maps API key (AK) and the application's signing certificate. See the FAQ for troubleshooting guidance.

7.4 Positioning Results Summary

The bottom information panel contains two rows.

First row: Displays the current latitude and longitude. Tap this row to expand the detailed results panel.

Second row: Displays five positioning indicators.

Indicator	Description

Horizontal Acc	Estimated horizontal positioning accuracy, in meters.
Vertical Acc	Estimated vertical positioning accuracy, in meters.
Speed	Doppler-derived speed, in km/h.
Sats	Number of satellites used in the positioning solution.
Mode	Current positioning mode (SPP / DGNSS / RTK), with the solution status indicated where available.

Interpretation Guidelines

- **Accuracy indicators represent estimated uncertainty derived from the positioning solution**, not independently measured positioning accuracy. The actual error may be either larger or smaller.
- **Satellite count:** SPP commonly uses approximately 15–30 satellites when sufficient signals are available. RTK may use fewer satellites because of double-difference processing and observation quality-control filtering.
- **RTK status:** When the mode is RTK, check whether the solution is fixed or float. A float solution generally has substantially lower accuracy than a correctly fixed solution.

7.5 Correction Source Configuration (NTRIP / CORS)

RTK and DGNSS require differential correction data. P-Nav supports connecting to network-based CORS services through NTRIP.

Configuration Steps

1. Obtain an account from a CORS service provider. The provider typically supplies the server address, port, mount point, username, and password.
2. Open **Settings → NTRIP** and enter the following information.

Field	Description
IP Address	Server address supplied by the service provider.
Port	Server port supplied by the provider. Common examples include 2101 and 8001.
Mount Point	Name of the correction stream. Tap **Refresh** to retrieve the available mount-point list from the server.
Username	Account username supplied by the provider.
Password	Account password supplied by the provider.

3. Return to the Positioning page and enable the **CORS** switch.
4. After a successful connection, check the status shown at the top of the page to confirm that correction data are being received.

GGA NMEA Settings in the lower part of the settings screen determines which position information is sent back to the server. Many CORS services require an NMEA GGA sentence to identify the approximate rover location and provide corrections for the appropriate region.

Option	Description
From positioning database	Uses the latest position available in the application's positioning database. This is the default option.
Manual input	Uses manually entered coordinates.
Phone GNSS location	Uses the location reported by the Android system location service.

NTRIP accounts are provided by individual CORS service providers. P-Nav does not include an account. Contact the relevant provider for account registration, fees, and mount-point selection. The server and mount point shown in the screenshot were used with a temporary test account and are provided only as examples of the configuration format.

7.6 Direct RTCM Connection

If a local base station is available, P-Nav can receive its RTCM data stream directly without an NTRIP account. Enable the **RTCM** switch and configure the connection parameters as required by the base station.

7.7 Ephemeris Status

The top of the page displays the number of loaded broadcast ephemeris records and the number of observations that have passed the relevant checks. These indicators help users verify whether the data required for positioning are available.

7.8 Positioning Configuration

Positioning parameters can be adjusted under **Settings → Solution Settings**. Changes take effect in subsequent positioning processing after the settings are saved.

The following figure shows the upper part of the settings page on the left and the lower part of the same page after scrolling on the right.

GNSS System

Select the constellations to be used in positioning: G (GPS), C (BeiDou), R (GLONASS), and E (Galileo). All systems are enabled by default. If a constellation provides poor-quality observations—for example, if its data completeness is very low on the Quality Analysis page—it may be disabled to prevent unreliable measurements from affecting the solution.

Atmosphere and Model

Parameter	Available Values	Description and Recommendations
Frequency	Single Freq / Dual Freq	Selects single-frequency or dual-frequency processing. Dual-frequency observations can be combined to form an ionosphere-free linear combination, which can reduce first-order ionospheric delay, but require the device to track two frequencies. The default is Single Freq.
Cut-off Angle	0°–90° in 5° increments	Elevation-angle mask. Satellites below the selected angle are excluded from positioning. The default is 15°. Increasing it to 20°–25° may help exclude low-elevation observations in environments with severe multipath, although fewer satellites will remain available.
C/N0 Threshold	0–45 dB-Hz in 5 dB increments	C/N_0 threshold for observation screening. Measurements below the threshold are excluded. The default is 0, which disables threshold-based screening. A threshold of 25–30 dB-Hz may help exclude weak signals.
Ionosphere Model	OFF / Klobuchar / Iono-Free LC	Selects the ionospheric correction method. **OFF** disables the model-based correction; **Klobuchar** applies the broadcast ionospheric model and is suitable for single-frequency processing; **Iono-Free LC** uses an ionosphere-free linear combination and requires dual-frequency observations.
Troposphere Model	OFF / Saastamoinen	Selects the tropospheric correction model. Saastamoinen is generally recommended for standalone positioning. Much of the tropospheric contribution is reduced by differencing in relative positioning, although residual effects may remain.
Stochastic Model	Height-Angle / SNR	Selects the observation weighting model. **Height-Angle** assigns weights according to satellite elevation, generally giving higher weights to high-elevation satellites. **SNR** uses signal-strength information for weighting and may be beneficial in challenging environments.

Note on the OFF settings: The built-in default configuration uses Klobuchar and Saastamoinen. The screenshot shows the currently saved configuration, in which both models are set to OFF. Such settings may be used in differential relative positioning, where differencing reduces much of the atmospheric error. For standalone SPP, enabling the appropriate ionospheric and tropospheric corrections is generally recommended because disabling them may degrade absolute positioning accuracy.

Ambiguity Resolution

Parameter	Available Values	Description

AR	OFF / Continuous / Instantaneous / Fix and Hold	Selects the integer ambiguity resolution strategy. **Continuous** (default) maintains ambiguity estimates across epochs and is the common choice. **Instantaneous** attempts ambiguity resolution independently at each epoch; it avoids relying on ambiguity continuity across epochs but can be more difficult to fix reliably. **Fix and Hold** applies constraints after ambiguities are fixed, which can accelerate stabilization but may propagate errors if an incorrect fix is accepted.
AR(R)	OFF / On / Auto Cal	Controls GLONASS ambiguity resolution. R is the RINEX system identifier for GLONASS. Because GLONASS uses frequency-division multiple access (FDMA) and can exhibit inter-frequency biases, ambiguity resolution is more challenging than for some other constellations. OFF is a conservative option.
Min Ratio to Fix Ambiguity	Numeric value	Sets the minimum ratio-test threshold for accepting an integer ambiguity solution. The ratio compares the residual sum of squares of the second-best candidate with that of the best candidate. A higher threshold imposes a stricter acceptance criterion. RTKLIB commonly uses **3.0**; values of 4.0–5.0 are stricter options that may reduce incorrect fixes at the expense of a lower fix rate.

BeiDou ambiguity resolution is enabled internally by default and does not have a separate switch on this page.

Output and Recording

Parameter	Available Values	Description
Solution Format	XYZ / BLH	Specifies the output coordinate format. **BLH** represents geodetic latitude, longitude, and ellipsoidal height and is the default. **XYZ** represents Earth-Centered, Earth-Fixed (ECEF) Cartesian coordinates, which are useful for post-processing and comparison with reference coordinates.

Tap **SAVE SETTINGS** at the bottom of the page to save the configuration. The settings are stored locally and remain effective after the application is restarted.

7.9 Detailed Solution Panel

Tap the information panel at the bottom of the Positioning page to expand the detailed solution panel. It provides more comprehensive positioning status information than the summary and helps diagnose abnormal processing.

Top Counters

Field	Description
Valid Satellites	Number of satellites in the current epoch that have both available observations and matching ephemerides.
Epoch Count	Cumulative number of positioning epochs processed during the current session.
RTCM Satellites	Number of satellites represented in the received real-time correction data. This value is nonzero only when RTCM or CORS correction data are being used and contain applicable satellite information.

Position

Field	Description
Latitude / Longitude	Current positioning solution, displayed in degrees, minutes, and seconds.
Altitude	Ellipsoidal height in the WGS 84 reference system.

If the solution is SPP and the estimated height remains noticeably different from the reference value, this may reflect the relatively poor vertical performance commonly observed in smartphone GNSS positioning. Vertical errors are often two to three times larger than horizontal errors, although the actual ratio varies with the device and observation conditions. P-Nav does not independently validate vertical positioning accuracy.

Accuracy (m)

The panel provides estimated accuracy values for the E, N, and U directions (east, north, and up), together with the **ratio** value.

The ratio is a key indicator for evaluating the reliability of RTK integer ambiguity resolution. It is defined as

$$\text{ratio} = \frac{\Omega_{\text{second-best}}}{\Omega_{\text{best}}} \tag{1.16}$$

where $\Omega_{\text{second-best}}$ and (Ω_{best}) are the residual sums of squares for the second-best and best integer ambiguity candidates, respectively. A larger ratio indicates greater separation between the two candidates and is generally considered evidence of a more reliable fix.

Ratio	Interpretation
< 3	Typically fails the ambiguity-resolution threshold; the solution remains float.
3–5	Meets a commonly used fixing threshold; reliability should be assessed with other indicators.
> 5	Generally indicates stronger separation between the best and second-best candidates.

= 0	Float solution; integer ambiguity fixing has not been performed or no valid ratio is available.

The threshold can be adjusted under **Min Ratio to Fix Ambiguity** in the positioning configuration settings. The default is 3.0.

Important: A high ratio does not guarantee that the ambiguities have been fixed correctly. Smartphone observations may occasionally produce an incorrect fix despite a high ratio. This can appear as a meter-level jump in the position time series while the ratio remains apparently acceptable. Therefore, evaluate fix quality by checking both the ratio and the continuity of the positioning results.

Satellite Information

This section lists the satellites used in the current positioning solution, together with their observation types, C/N_0 values, and roles in processing. It can help users identify which satellites were excluded and investigate possible reasons for their exclusion.

Solution Log

The solution log displays epoch-by-epoch processing messages from the RTKLIB engine. It generally does not require attention during normal operation, but provides useful diagnostic information when positioning fails or behaves unexpectedly.

8. Ephemeris Download

GNSS positioning requires satellite orbit information. P-Nav supports two types of ephemeris products:

Type	Purpose	Files
Broadcast ephemeris	Default source for SPP, DGNSS, and RTK	Mixed BRDM navigation files (RINEX 3) or constellation-specific files
Precise products	Improve orbit and clock accuracy for SPP	SP3 orbit files and CLK clock correction files

8.1 Data Sources

P-Nav has three built-in FTP servers and switches between them automatically in the following order:

Priority	Server	Description
1	igs.gnsswhu.cn	Wuhan University IGS Data Center; preferred for faster access from China
2	igs.ensg.ign.fr	IGN, France
3	isdcftp.gfz-potsdam.de	GFZ, Germany

If a server times out or the requested file is unavailable, P-Nav automatically tries the next server. During downloading, the interface displays the file name and size. Downloaded .gz files are automatically decompressed, and outdated files are cleaned up. Files for the current and previous days are retained.

8.2 Usage Notes

- **Broadcast ephemeris requires an internet connection to download the current day's data.** It is recommended to complete the download before starting a measurement session.
- **Precise products are released with a delay.** Ultra-rapid products may become available several hours after the relevant observation time, so they are often unavailable for real-time measurements and are more suitable for post-processing.
- The download interface lists files already present on the device. Select the files you want to delete directly from this list.

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9. Data Export

P-Nav provides three types of exported data, each serving a different purpose:

Export Type	Contents	Typical Use
RINEX 3.04 observation files	Raw GNSS observations	Post-processing with RTKLIB, Bernese, GAMIT, and other tools
GnssLogger-compatible CSV	Raw observations and positioning results	Inspecting data with analysis tools in the GnssLogger ecosystem
Track export (KML / GPX / CSV)	Positioning results	Map visualization, deliverable generation, and data exchange between applications

The first two types are generated automatically when data collection stops. Track export must be triggered manually.

9.1 RINEX 3.04 Observation Files

RINEX observation files are generated automatically when data collection stops. No additional operation is required.

File naming convention:

PNAV<day-of-year><session-letter>.<two-digit-year>o

For example, PNAV278e.26o represents session *e* on day 278 of 2026. When multiple sessions are recorded on the same day, the session letters increment sequentially (a, b, c, and so on).

File structure:

- **Header:** RINEX version, station and receiver information, approximate position for post-processing initialization, observation-type declarations, and carrier-phase offsets for individual frequencies.
- **Observation data:** Measurements recorded at each epoch, with one line per satellite and satellites organized by constellation.

RINEX observation-type adaptation: The observation types declared in the file header depend on the frequencies actually observed during the recording session. **Only frequencies for which observations were recorded are declared.** Consequently, files generated by the same phone model may contain different observation-type lists under different observation conditions.

Observation codes:

Constellation	Frequency	Observation Code
BDS	B1I	2I
BDS	B1C	1P
BDS	B2b	7Z
BDS	B2a	5P

The observation-type prefixes indicate the measurement type: C for pseudorange, L for carrier phase, D for Doppler, and S for signal strength.

The valid tracking codes for BDS B2a include 5D, 5P, and 5X. The current version of P-Nav writes B2a observations using 5P.

9.2 GnssLogger-Compatible CSV

A GnssLogger-compatible CSV file is generated alongside the RINEX file when data collection stops. It uses a text-based record structure consistent with Google GnssLogger.

The file contains five record types:

Record Type	Contents
Raw	Raw observations, including time, carrier phase, pseudorange, Doppler, carrier-to-noise density ratio (C/N_0), and measurement state flags
Fix	Positioning results, including latitude, longitude, altitude, velocity, and accuracy
NMEA	Original NMEA sentences
Nav	Satellite navigation messages, if supported by the device
Status	Satellite status, including constellation, azimuth, elevation, and whether the satellite was used in the positioning solution

Lines beginning with # define the record columns. This format can be read directly by Google's GNSS Analysis App and other tools in the GnssLogger ecosystem.

9.3 Track Export (KML / GPX / CSV)

Track export converts recorded **positioning results** into common geospatial formats for map visualization and deliverable generation.

How to export: Open the export dialog from the file list or the Positioning page, and select the desired format. P-Nav generates the file and opens the **system share sheet**, allowing you to send it through WeChat or email, upload it to cloud storage, or save it locally.

Differences between formats:

Format	Contents	Typical Applications
KML	Three-dimensional track, including the route and individual positions	Google Earth, OvitalMap, and similar map applications
GPX	Track points with timestamps	Outdoor navigation software, activity-tracking platforms, and many GIS tools
CSV	Tabular positioning records	Spreadsheet analysis and custom post-processing

Exported content: Track exports contain positioning results only; they **do not include raw GNSS observations**. To post-process raw observations, use the RINEX files described in Section 9.1.

File naming convention:

pnav_track_<timestamp>.kml

pnav_track_<timestamp>.gpx

pnav_data_<timestamp>.csv

The timestamp indicates the time of export. Exporting the same dataset multiple times therefore generates separate filenames rather than overwriting the previous export.

9.4 File Locations

RINEX observation files and GnssLogger CSV files are stored in the application's private external-files directory:

/Android/data/com.ecut.pnav/files/PNav/

These files can be accessed through a USB connection, a file manager, or the **Files ›** entry within the application. Application runtime logs are stored in the PNav/logs/ subdirectory.

Track export files are first written to a temporary directory and then passed to the system share sheet. If you do not select a sharing or saving destination, the files may not be retained permanently. To archive an export, select **Save to Files** or another explicit destination.

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10. Frequently Asked Questions

Q: Why does the epoch count stop increasing after I start data collection?

Check whether location permission has been granted. On some devices, you may also need to enable **Precise Location** in the system settings.

Q: Why is positioning accuracy poor even though many satellites are visible?

Check the PDOP value. If satellites are concentrated in one part of the sky, the satellite geometry may be poor, resulting in a higher PDOP. You can also inspect the Quality Analysis page to identify satellites with low C/N₀ that may indicate degraded observation quality.

Q: Why does RTK fail to achieve a fixed solution, or why does the ambiguity ratio remain below the threshold?

Common causes include:

1. No valid correction data is being received.
2. C/N₀ is low or multipath effects are severe.
3. The phone tracks L1+L5 rather than the L1+L2 combinations commonly used by surveying-grade receivers. Inter-frequency biases affecting L5 observations may make ambiguity resolution more difficult.
4. Frequent cycle slips occur in dynamic environments.

Start by checking the observation conditions on the Quality Analysis page.

Q: Why does the application take a long time to respond after I stop data collection?

Observation data are buffered in memory during collection and written to files when collection stops. Longer sessions therefore require more time to process and save. Wait for the operation to complete and avoid force-stopping the application.

Q: Why can another software package not read the BDS observations in an exported RINEX file?

First, verify that the target software supports the RINEX version and observation codes used in the file. If the file was generated by an older version that encoded B2a observations as C5Q, try exporting the data again using the current version, which writes the B2a code as C5P.

Q: Why is the map not displayed?

The Baidu Maps SDK associates its application key (AK) with the signing certificate. A build signed with a different certificate may require its own AK. See app/libs/README.md in the project repository for details.